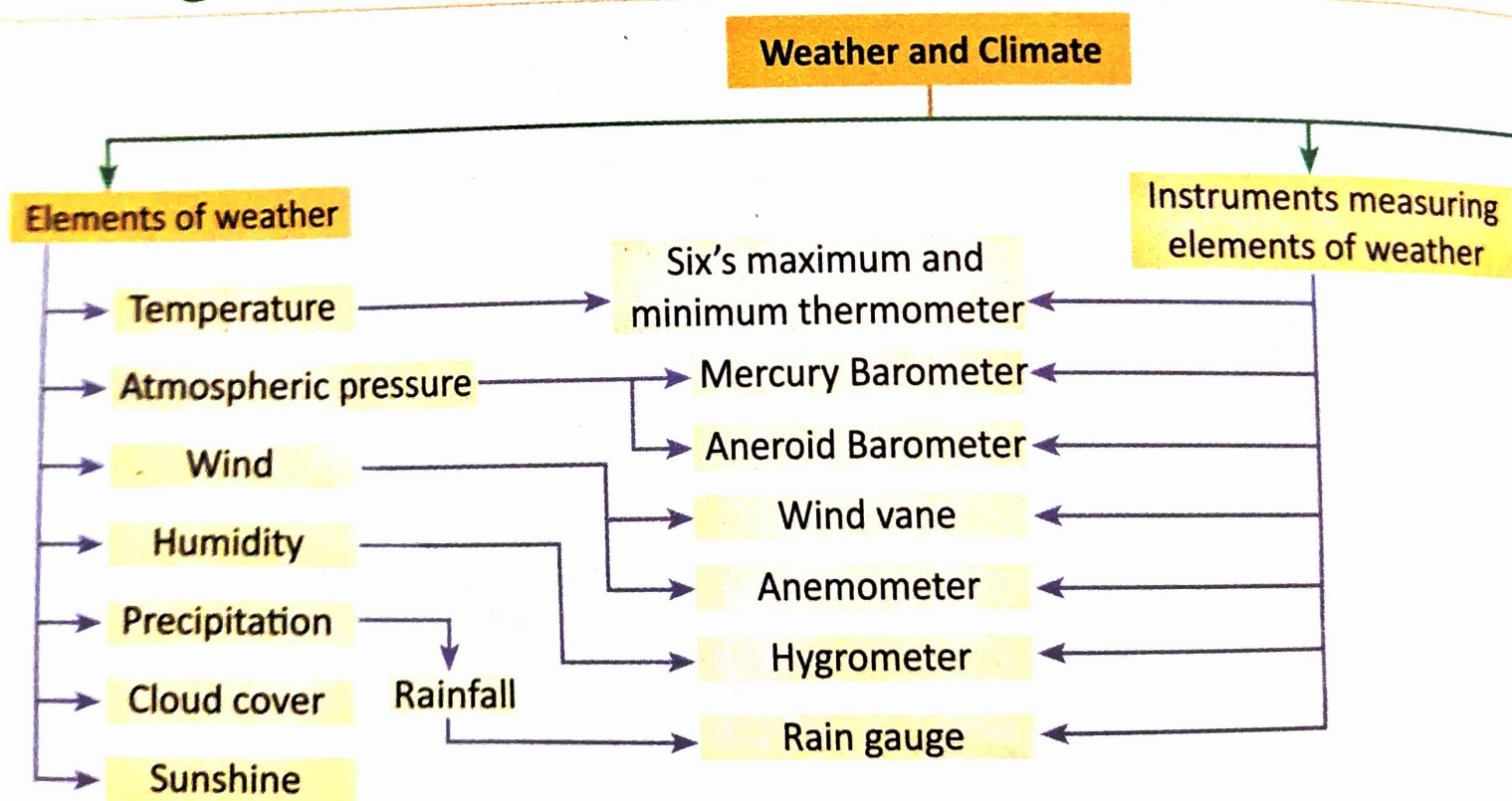


Weather and climate are major determinants of how we lead our lives. The dress we wear, the food we eat, and even the kind of shelter we have depend on the weather and climate of the place we live in.

- What exactly do weather and climate mean? How are they different?
- What are the major elements of weather and climate?

During Reading



Warm-Up

This is a picture of *Kalbaisakhi*, a local thunderstorm that blows in West Bengal. The word means 'calamity in the month of Baisakh'.

Find out the names of three similar local thunderstorms in India. Write their names in the blanks provided:

1. _____
2. _____
3. _____

21st
CENTURY
SKILLS

- information literacy
- media literacy
- critical thinking



WEATHER AND CLIMATE

Not all days of a year feel the same. Some days are hot, some days make us feel cold, some days are cloudy, while some days are humid and make us sweat a lot. Sometimes, days are marked by bright sunshine, while some days may be wet and windy. Daily change in the state of the atmosphere of a place is known as **weather**.

Weather is defined as the atmospheric condition of a certain place at a particular point of time. All weather changes take place in the lowest layer of the atmosphere—the troposphere.

→ Due to the presence of water vapour.

There may be different weather conditions even during the course of a day. A sunny morning may be followed by a rainy afternoon. Again, two regions within the same city may experience different weather conditions. For example, Colaba in south Mumbai may experience heavy showers, while Juhu, situated slightly in the north, may enjoy a sunny day at the same time. Weather changes over a short period of time.

Climate, on the other hand, refers to the sum total of the average weather conditions that prevail at a particular place over a long

period of time. Daily weather conditions such as temperature, rainfall, humidity, cloud cover and sunshine of a place are analysed over a period of 30 to 35 years to determine the climate of the place. Accordingly, we can say the Sahara Desert has a hot desert type of climate, while Sri Lanka has an equatorial

Weather	Climate
Weather refers to the conditions of the atmosphere of a particular place over a short period of time.	Climate is the average condition of the atmosphere calculated over a long period of 35 years or more.
It is highly dynamic. It may change very quickly.	It is more stable and permanent in nature.
Weather is determined over a small area.	Climatic data are calculated for a large area such as a state or a country.
Weather influences the daily activities of the people over a local area.	Climate helps to determine and understand broader themes such as climatic belts, natural vegetation regions and cropping patterns of a country or continent or the world as a whole.

climate. Several other types of climate are prevalent throughout the world, each having its own unique characteristics.

Thus, we see that while weather may behave in an erratic manner, climate is far steadier and also covers a larger area. Earlier, to determine the climate of a place, observation and recordings of different elements of weather such as temperature, pressure and

precipitation were taken. Now, satellite images provide relevant information the climate of a region more quickly and accurately.

ELEMENTS OF WEATHER

Weather has several components or elements that are essential for determining the climate of a region or a place.

KEY WORD

satellite images: pictures of Earth taken from space